AWS Cloud Compute Grant Proposal

OpenCV AI Competition 2026, powered by AWS

Team Name

Steropes devs

Project Title

Physically-Grounded Model Cascade

Problem Statement & Real-World Impact

Continuous, always-on visual monitoring systems face a hard cost-accuracy trade-off. Lightweight local models are fast and free to run but fail silently on ambiguous, occluded, or unfamiliar scenes precisely the moments where getting it right matters most. Powerful vision-language models (VLMs) are far more reliable on hard cases, but they are too slow and too expensive to run on every frame of a continuous stream.

Existing model-cascade research chooses between local and cloud inference purely in software, based on a confidence score. None of it considers that a physical camera can simply move reposition, re-angle, or get closer to gather better evidence before paying for an expensive cloud call. This is especially costly in remote deployments where cloud connectivity is metered, such as environmental or agricultural monitoring stations relying on satellite (Iridium-class) uplink, where every cloud query has a real, non-trivial dollar cost, while physical repositioning only costs a small amount of battery.

Impact: for every ambiguous frame the system resolves through physical repositioning alone, it avoids an expensive, bandwidth-constrained cloud call entirely. Over weeks or months of continuous deployment, this translates into a quantifiable, real operating-cost reduction while preserving cloud-level accuracy on the genuinely hard cases motion alone cannot resolve.

Target Users & Beneficiaries

- Conservation and wildlife research groups running remote camera-trap style monitoring
- Remote agricultural operations monitoring crops, livestock, or equipment
- Infrastructure operators in areas with limited or metered terrestrial/satellite connectivity

Planned OpenCV 5 Image / Video Analysis

- Tier 1 local detection via cv2.dnn, running OpenCV 5's native inference engine on the edge device
- Confidence/uncertainty scoring combining detection confidence, multi-object tracking consistency, and optical-flow stability across frames
- Re-run of the same detection pipeline after physical repositioning, so OpenCV 5 performs real, repeated, load-bearing work at every tier
- Camera pose/calibration bookkeeping to reconcile observations taken from different viewpoints

Planned AWS Architecture & Services

Layer	Service	Role
Edge compute	Raspberry Pi 5 / Jetson Orin Nano	Runs Tier 1 + Tier 2 detection locally; most frames resolved with zero cloud involvement.
Cloud escalation	Amazon Bedrock (VLM)	Invoked only for Tier 3, the expensive last-resort path.
Agent / decision logic	AWS Lambda / lightweight agent runtime	Hosts the escalation decision logic and adaptive-threshold learning loop.
Communication bridge	AWS IoT Core (MQTT)	Carries reposition commands to hardware and execution acknowledgements back.
Cost / state tracking	DynamoDB	Logs every decision: tier resolution, confidence values, reposition and escalation events.
Parallel processing	ECS Fargate on Graviton (ARM64)	Batch reprocessing and benchmarking workloads.
Frontend hosting	S3 + CloudFront	Public judge-facing dashboard, no login required.

High-Level Architecture / Technical Description

Frame captured → Tier 1 local detection (OpenCV 5 / cv2.dnn) with confidence scoring. If confidence is high, the frame is resolved with no further cost. If confidence is low, the AI Agent issues a physical reposition command (pan/tilt/approach) over AWS IoT Core / MQTT; the hardware controller executes the move and a new frame is captured and re-evaluated. If confidence is now high, the frame is resolved with the cloud never contacted. If confidence remains low, the Agent escalates to Amazon Bedrock (Tier 3) the only path that costs real connectivity budget. The result is logged to DynamoDB, and the adaptive confidence threshold is updated based on whether the escalation was actually necessary.

The Agent decides WHAT action should happen; the Hardware layer decides HOW that action is physically executed; AWS IoT Core / MQTT is the communication bridge between them, carrying structured JSON commands and acknowledgements.

Proposed Evaluation Method & Judge Demonstration

- Tier-resolution rate: across a batch of test frames of varied difficulty (clear, occluded, glare, bad angle), report the fraction resolved at each tier.
- Physical repositioning effectiveness: detection accuracy before vs. after Tier 2 reposition on a fixed set of intentionally ambiguous starting angles.
- Cloud-call reduction with an honest cost estimate: Tier 3 escalations avoided vs. a local-only / cloud-always baseline, translated into a concrete dollar figure under a stated connectivity-cost assumption.
- Adaptive threshold validity: threshold changes over a test session and resulting change in escalation accuracy (necessary vs. unnecessary).
- Documented failure cases: scenes where even repositioning fails to resolve ambiguity (heavy occlusion, extreme lighting, novel objects), honestly reported.

Judge demonstration: a live dashboard (S3 + CloudFront) showing, per frame, which tier resolved it, a running cost-savings counter, adaptive-threshold visualization, and a decision/event timeline, alongside a live or recorded demonstration of the physical repositioning hardware.

Featured Path

Agentic Vision Award. The system implements a genuine perception-decision-action loop: low confidence directly triggers a physical motor command, and the resulting new observation directly determines whether an expensive cloud call happens at all visual understanding changes a subsequent action, not just an explanation.

Team Bio

Name	Role
Rahma shahbaz	AI / Computer Vision Engineer (OpenCV 5)
Waad Daraghmeh	Backend Engineer
[Name]	Frontend Engineer
Basel ayoub	Hardware / Robotics Engineer
Mu’ad ahmed hassan	AI Agent / AWS Engineer